

IN-CLASS ACTIVITY

Your Bike

Six questions about form, asked of something you already know

Group members: _____

Date: _____

Reading: Chapter 4 of [Crawley 2015](#) systems.

Scenario

You ride a bike to campus. It is nothing special: a steel frame, one gear, brakes that squeeze the wheel rims, and a rack over the back wheel that you added last year.

Three things happened this month.

1. The rear brake started rubbing. Nothing is broken, nothing came loose, and every bolt and cable is exactly where it has always been.
2. You ordered a new rack online. It arrived the right size, and it will not go on. Your frame has no bolt holes where the rack expects them.
3. You transferred here from a campus that was flat. The bike has not changed in any way. It is now a much worse bike.

What You Are Doing

Chapter 4 asks six questions about the form of a system. They are collected in its Table 4.1. You are going to answer all six about your bike.

Chapter 4 is about what a system **is**. What it **does** is the next chapter. So when you describe the bike, say nothing about what its parts are for, and notice how hard that is when every part of a bicycle is named after its job. The one place function comes in is the definition of form itself, which says form is *instrumental* in function. You will need that in Part 2b.

Short answers. A sketch usually beats a paragraph.

1 What is the system?

Reminder: the definition of form (Box 4.1)

Form is the physical or informational embodiment of a system that **exists** or has the potential for stable, unconditional existence, for some period of time, and is **instrumental in the execution of function**. Form includes the entities of form and the formal relationships among the entities. Form **exists prior to the execution of function**.

1a. Warm-up: say it without function words

The text opens on how hard this is. We describe a coffee cup with “handle”, a pencil with “eraser”, a notebook with “binding”, and every one of those words is a purpose wearing a noun. The form-only versions are “flat cardboard half-circle”, “rubber cylinder”, and “metal spiral”.

Function word	Form-only description
Brake	
Seat	
Handlebar	

1b. The abstraction and its specialization

Worked example (Figure 4.6 of the text)

For the centrifugal pump, the text creates the abstraction **Pump**, which “brings to the surface the idea of something that moves fluid but also hides all details of motors, impeller, and the like.” It then hangs the more specific **Centrifugal pump** off it with a specialization triangle.

Your turn.

General abstraction: _____

Specialization: _____

2 What are the principal elements of form?

2a. Write the parts list

The text starts its pump analysis from a parts list, and Figure 4.5 simply hands it to you with the parts numbered. Your list starts with five parts given to you here, because Part 3 is built on them. The descriptions stick to form.

Five parts you are given

- **Frame:** welded steel tubes forming two triangles. The rear wheel and the brake arms mount directly on it; the handlebar attaches through the fork and stem.
- **Rear wheel:** a metal rim laced by spokes to a central hub. The hub's axle is clamped into two slots at the back of the frame.
- **Brake pad:** a rubber block bolted to a metal arm on the frame. It sits about two millimeters from the side of the rear rim and does not touch it.
- **Brake lever:** a pivoting metal lever clamped to the handlebar. A steel cable runs from it to the arm that holds the pad.
- **Chain:** a loop of steel links wrapped around a toothed ring at the pedals and a smaller toothed ring on the rear hub. It runs close beside a frame tube without touching it.

Now add the rest. Name whatever other parts you can, in any order. A dozen is plenty, the list does not have to be complete, and nothing later in the hour depends on getting all of it.

2b. Three things that are also on the bike

Also present

the air in the tires • the mud on the frame • the rider

Box 4.1 gives three tests: the thing exists for some period of time, it is instrumental in the execution of function, and it exists before the function is executed. Apply all three to each item. They do not all come out the same way.

Item	Form? Yes / No	Which test settles it, if any
The air in the tires		
The mud on the frame		
The rider		

Now two follow-ups. They need one idea from Chapter 2, the operand.

Reminder: form and operand

Function is a process acting on an operand. “The operand is the thing whose state is changed by that process” (Section 2.3). An object that exists but is not instrumental, or does not exist beforehand, fails the two extra tests of form, and the text says such objects “are operands, as discussed in Chapter 5” (Section 4.2).

Follow-up 1: the water and the air. The text’s pump is full of water whenever it runs, and yet water appears nowhere on its parts list (Figure 4.5, Table 4.2). The reason is that the water is the pump’s operand: raising the water’s pressure is what the pump is for (Section 4.3). Your tires are full of air. It is also a fluid, under pressure, inside a container. While you ride, does the air play the role the pump’s water plays, or a different one? Name the difference between the two cases that decides it.

Follow-up 2: same air, different process. Keep the same air in the same tire, but change the process. Name something you could do with the bike in which the air is the operand, the thing whose state the process changes. In that process, what is the instrument?

2c. Level-1 abstractions

Worked example (Figure 4.8 of the text)

The pump has ten elements of form. The text groups them into two Level-1 abstractions, **pump assembly** and **motor assembly**, then points out that the grouping is not unique: **rotating** and **non-rotating components** would have worked just as well.

Your turn.

Group your parts list into four to six Level-1 abstractions, and draw the two-level tree. Hang three or four parts under each.

Two short ones:

1. Which entries on your list are **classes** with more than one instance? Take the wheels. When is it honest to treat the two of them as a single class, and when does that stop being honest?
2. The text's alternative grouping for the pump was rotating against non-rotating. Try it on your bike. Does it work, and does it show you anything your first grouping hid?

2d. Hierarchy

Worked example (Table 4.3 of the text)

A hierarchy ranks elements by “scope, importance, performance, responsibility, or function” (Section 4.3), not by size or cost. The text puts five pump parts at the top and gives a reason for each: the cover and housing contain the fluid and provide the interfaces, the impeller does the work on the fluid, the motor drives the shaft and supports the housing, and the shaft drives the impeller. The fasteners (screws and locking nut) are lowest, since all they do is hold other parts in place. The O-ring, seal, and water slinger, which serve the top-grade parts, fill the middle. The point of the table is to tell you which five things to understand before all the rest.

Your turn.

Grade the parts from your list the same way, and write down the rule you used. Put individual parts in the table, not the Level-1 groups from 2c.

Grade	Elements
Highest	
Middle	
Lowest	

Your rule: _____

Now look at your *bottom* row, and at any small parts you left off the table altogether, such as bolts. Which low-grade item’s failure would hurt you the most? Given that answer, say in one line what hierarchy is good for, and what it is not.

3 What is the formal structure?

Reminder: two kinds of structure

Spatial and topological relationships say *where* things are: above, aligned, concentric, near, inside, touching. Location only.

Connectivity relationships say *what is joined to what*: bolted, welded, glued, cabled, meshed. They create the ability to transfer something.

Two things can be adjacent and not connected, and two can be connected with no particular location. The screening question for both: *is this relationship key to some important functional interaction or to the successful emergence of function and performance?*

3a. Fill in the DSM

Use the five given parts from 2a, and their descriptions. For each pair, read down the column to the row. Write **S**, **C**, or **SC**, and leave a cell blank if neither one matters. This is the same format as Table 4.7 in the text, which does it for the pump.

	Frame	Rear wheel	Brake pad	Brake lever	Chain
Frame	X				
Rear wheel		X			
Brake pad			X		
Brake lever				X	
Chain					X

3b. The cells that carry the lesson

Circle every cell where you wrote **S** alone and every cell where you wrote **C** alone. If you have none of either kind, go back over the brake pad's row before going on. Then answer this:

Why can you loosen the brake lever and slide it a few centimeters along the handlebar with no harm done, but you cannot move the brake pad two millimeters?

3c. The rubbing brake

Go back to the first incident in the scenario: your rear brake pad has started rubbing against the rim. Nothing is broken, nothing came loose, and every bolt and cable is where it has always been, so no connectivity relationship changed. Which kind of relationship did change? And what does that tell you about the exploded parts diagram in the back of a repair manual, where every part is drawn pulled away from its neighbors?

4 What is the whole product system, and where are the boundaries?

Reminder: two levels out

The **product/system** is what you are responsible for. The **accompanying systems** are the things outside it that are *essential* for value to be delivered. The two together are the **whole product system**. The **system boundary** runs between them, and every crossing of it is an **interface**.

One step further out is the **use context**: things normally present when the system operates that are *not* essential to delivering value. Use context establishes place, informs function, and influences design, even though none of it is required.

Candidates

the rider • the road • the air in the tires • a lock • lights • a helmet • a pump • the bike rack outside the building • the weather • the hills • other traffic • the campus bike registry

4a. Sort them

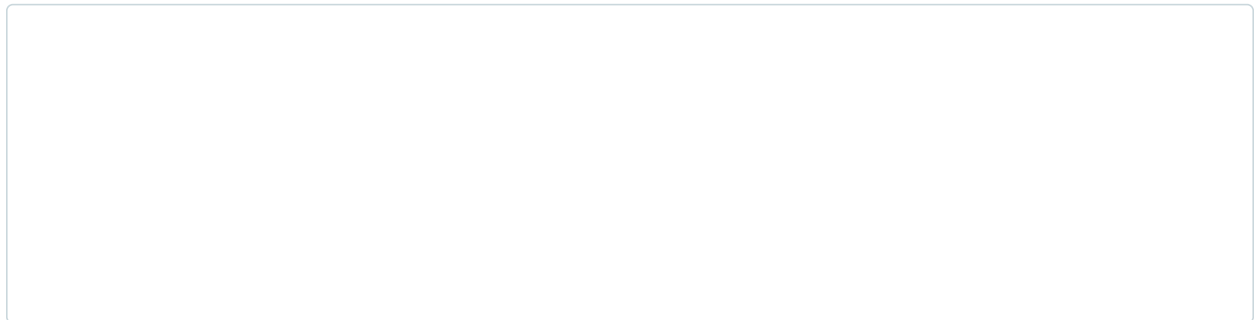
The test for the middle column is: *is this essential to the delivery of value?* If the answer is no but the thing is normally there anyway, it goes on the right.

Product / system	Accompanying system	Use context
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The bike is registered to you at the campus bike office. Section 4.4 of the text lists that kind of relationship alongside location and connection. Name it, and say whether it would appear on any drawing of the bike.

4b. Draw it, then list the interfaces

Draw the whole product system the way Figures 4.17 and 4.18 do, with a dashed product/system boundary. Then list every place a relationship crosses the line.



Interfaces:

4c. The rack that will not go on

This is the second incident: the new rack is the right size and was sold as described, but your frame has no bolt holes where the rack expects them, so nothing is defective and still it will not go on. Which interface is this, and which kinds of relationship does it involve? Now apply the text's remark in Section 4.5 about a pump and a third-party inflow hose: who ends up blamed, and is that fair? Finally, look at how your *old* rack is attached. What does that suggest you can do?

4d. The hills

The third incident: you moved from a flat campus to a hilly one, and the same bike is now much worse. Nothing about the bike changed. Not one element of form, not one relationship. By the value test, the hills are use context and not even part of the whole product system. In a sentence or two, what does that say about how far out an architect has to look, and why does the text put the number at two?